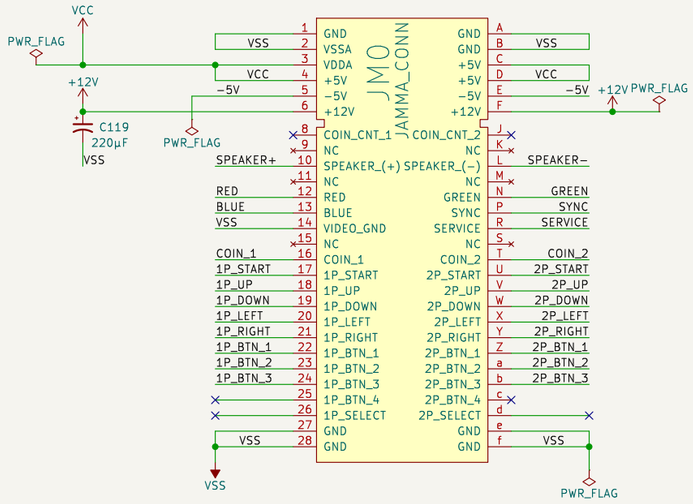
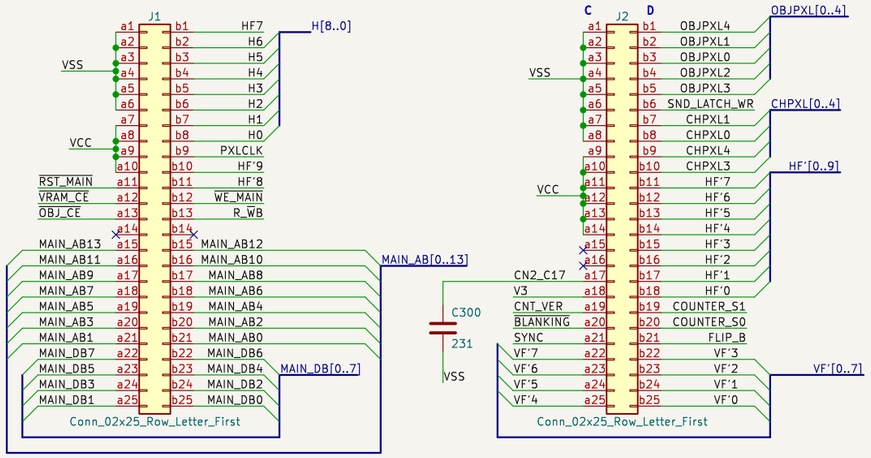
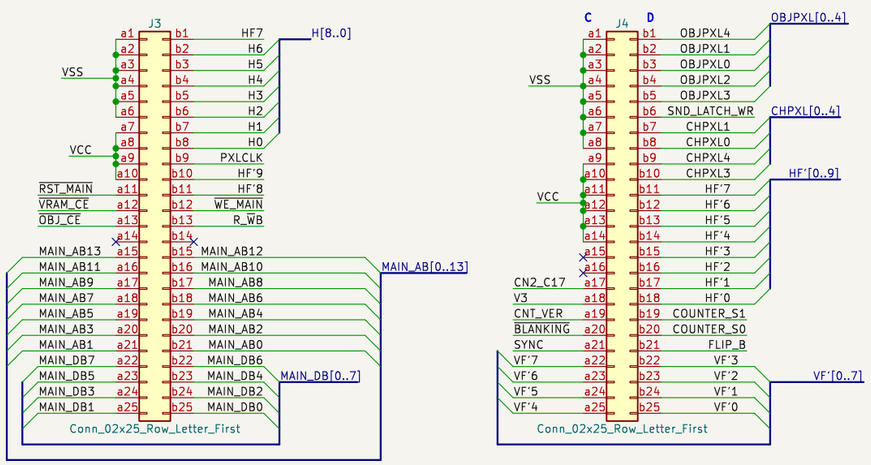
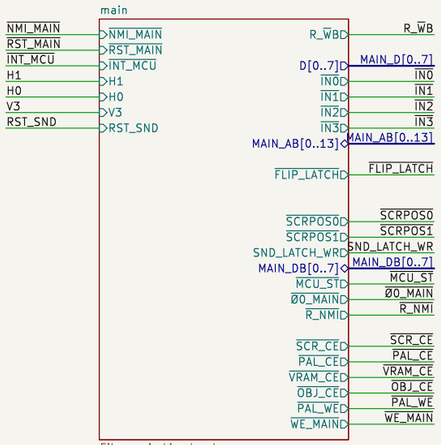


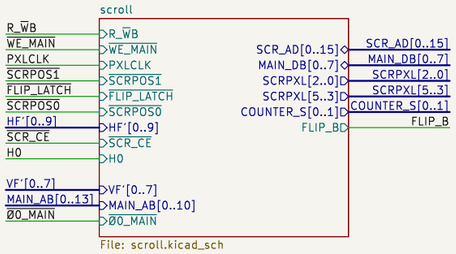
Top board connectors

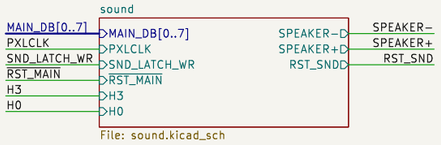


Bottom board connectors

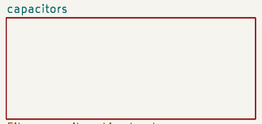


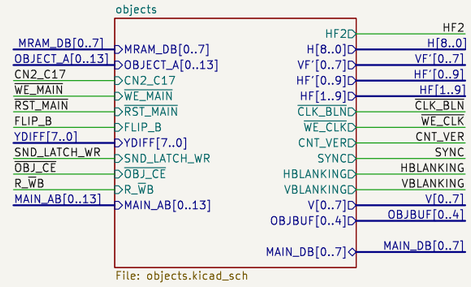


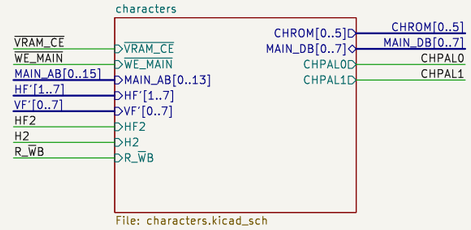


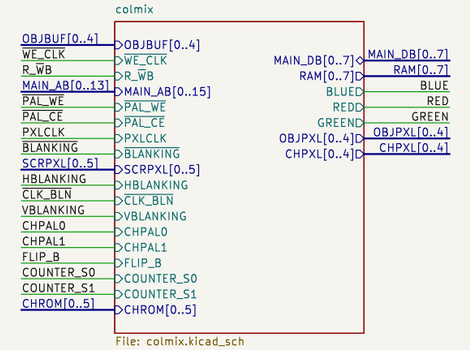


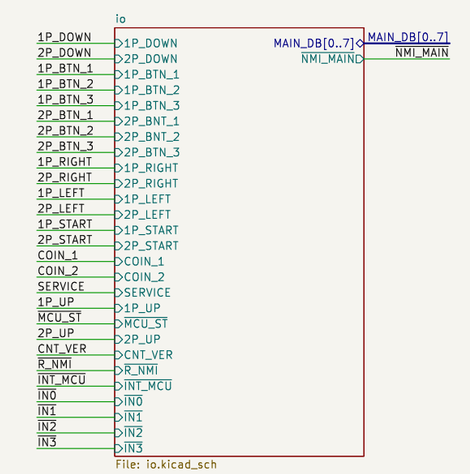












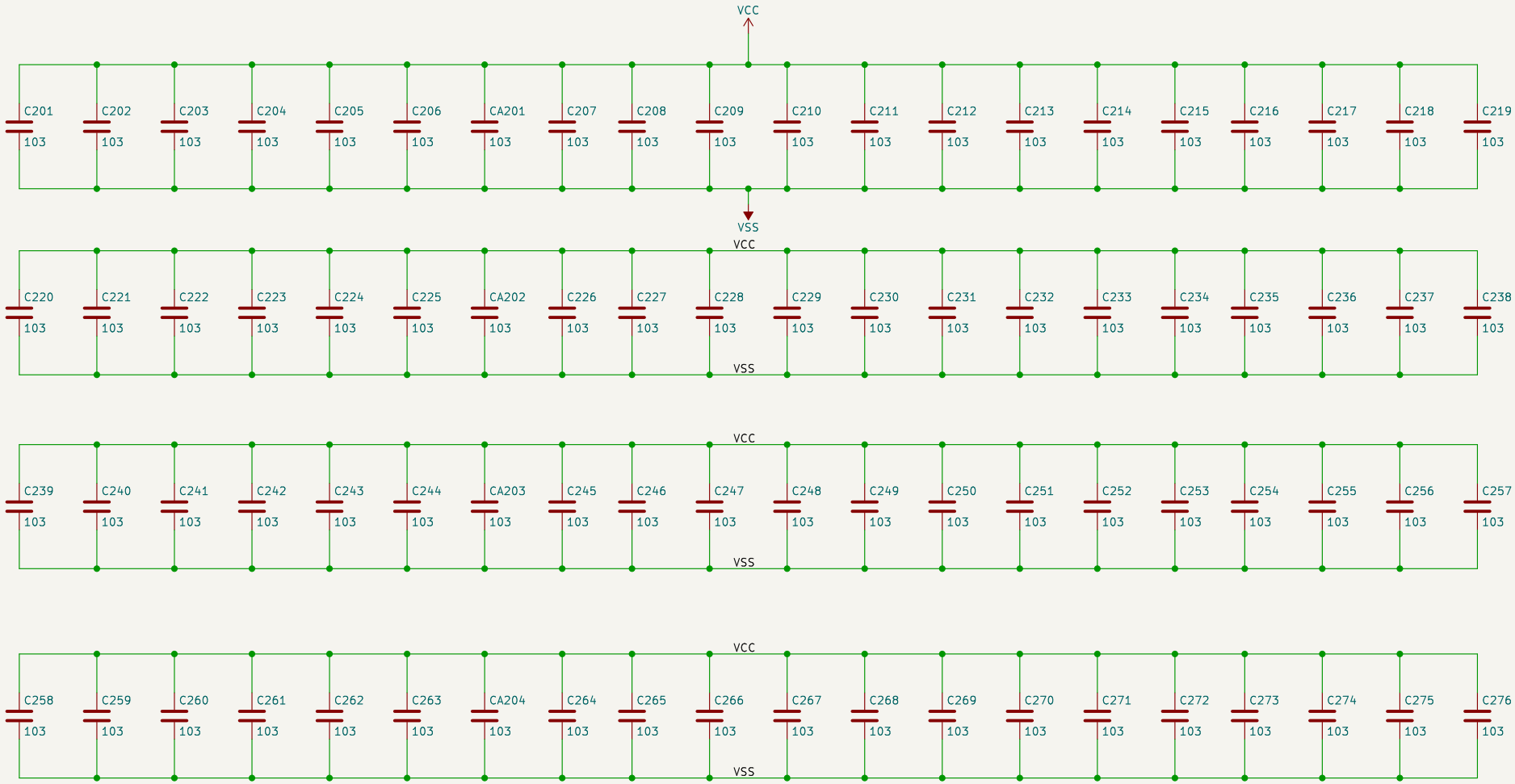
For repair and maintenance

JOTEGO

Sheet: /
File: kunio.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 1/16



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JOTEGO

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Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 2/16

SND_D[0..7]

MAIN_DB[0..7]

SND_A[0..14]

A[14..0]

D[7..0]

VCC

VSS

RESET

R49 4.7kΩ

R41 1kΩ

R42 47Ω

C96 4.7µF

F15D 74LS14

F15C 74LS14

RST_SND

A6 3-state flip flops 74LS374

SND_LATCH_WR

SND_LATCH_RD

74LS74

B5 MC68A09

R47 30KΩ

FIRQ

B1 HM65256B_SERIES 32kB RAM

A1 27256

D6B 74LS04

D6A 74LS04

E_SND

RW_SND

74LS00 A5B

WE

74LS138 AAA4 address decoder

A4A 74LS08

A4C 74LS08

RAM_CE

PCM_CTRL

PCM_START

PCM_STOP

RST_MAIN

D8F 74LS04

F2 YM3812-3526 FM OPL

H0

3.58MHz

74LS174 hex flip flop

E7C 74LS32

Y3 YM3014

G3B L8410-LM358N

G3A L8410-LM358N

R46 10Ω

C97 4.7µF

M9012-358N

L8410-LM358N

+12V

C100 100µF

MB3730

BYPASS

FILTER

H1

IN

OUT_M

OUT_I

INV. AMP

SPEAKER+

SPEAKER-

C99 10µF

NOT POPULATED

C101 4.7µF

C102 4.7µF

C103 220µF

R44 2.2kΩ

R45 4.7KΩ

RV1 R_Pot_Trim_US

C98 4.7µF

D1A 74LS393

D1B 74LS393

D3A 74LS393

D3B 74LS393

VCK

E1C 74LS04

PCM_RST

C1

C3

C4

74LS157 S Quad Mux

H3D

E2 OKI-M5205 Semiconductor converter 32kHz

R43 1Ω

C104 103

resistor values likely to be wrong

R48 1Ω

RN3 R_Network08_US

D4B 74LS74

NMI

For repair and maintenance

JOTEGO

Sheet: /sound/

File: sound.kicad_sch

Title: RENEGADE

Size: A3 | Date: 2022-07-08 | Rev: José Tejada

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MAIN_AB[0..13]

MAIN_A[0..15]

MAIN_DB[0..7]

PA[7..0]

MAIN2MCU

VCC

VSS

D[0..7]

Rota
No se puedo volcar

Volcada

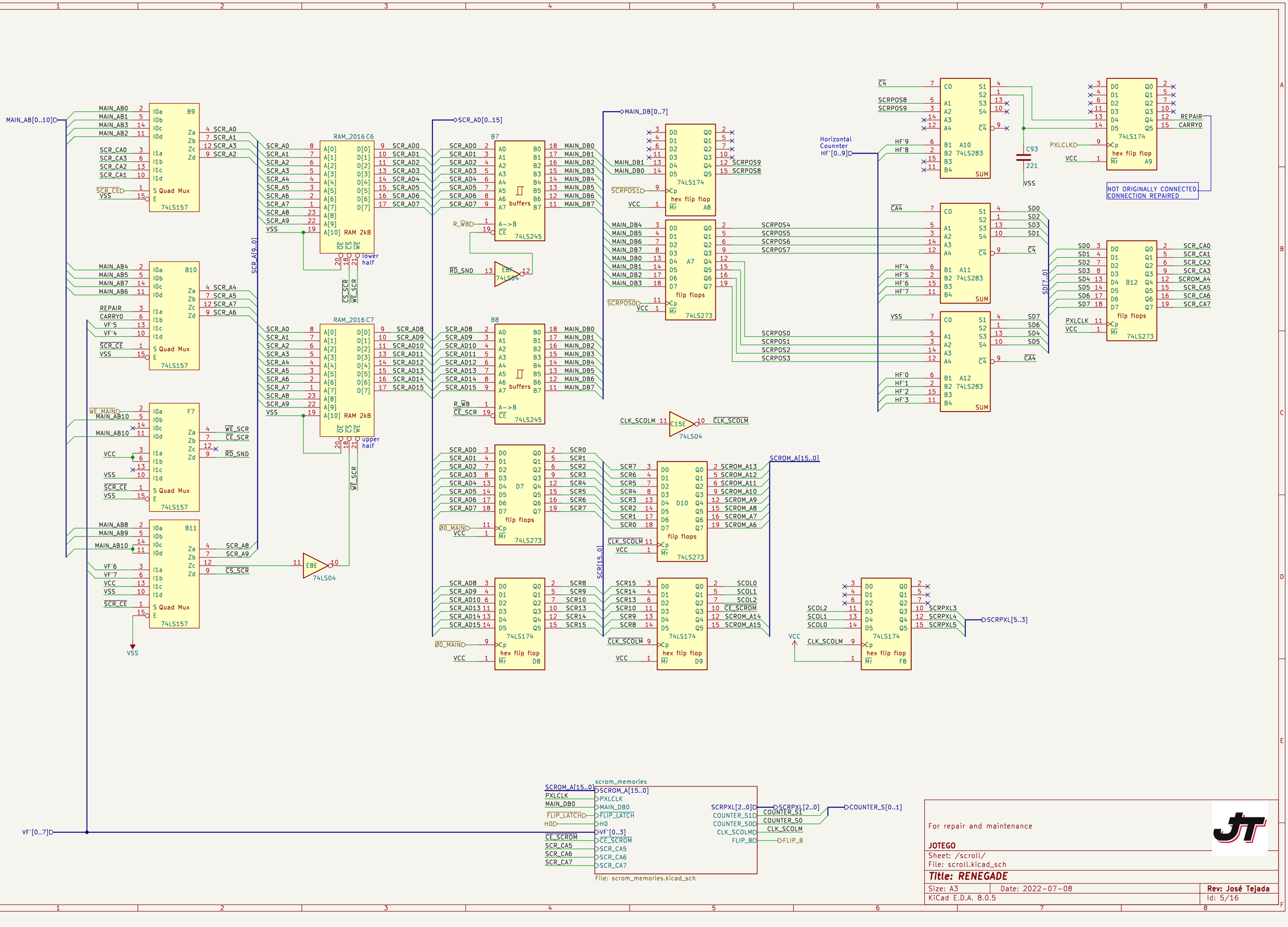
For repair and maintenance

JOTEGO

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Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 4/16


MAIN_AB[0..10]
MAIN_AB0
MAIN_AB1
MAIN_AB3
MAIN_AB2
SCR_CA0
SCR_CA3
SCR_CA2
SCR_CA1
SCR_CE
VSS
I0a
I0b
I0c
I0d
I1a
I1b
I1c
I1d
S Quad Mux
E
74LS157
B9
Za
Zb
Zc
Zd
SCR_A0
SCR_A1
SCR_A3
SCR_A2
SCR_A[9..0]
RAM_2016 C6
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10] RAM 2kB
D[0]
D[1]
D[2]
D[3]
D[4]
D[5]
D[6]
D[7]
SCR_AD0
SCR_AD1
SCR_AD2
SCR_AD3
SCR_AD4
SCR_AD5
SCR_AD6
SCR_AD7
OE
CS
WE
lower half
CS_SCR
WE_SCR
SCR_AD[0..15]
B7
buffers
A->B
CE
74LS245
R_WB
RD_SND
74LS04
MAIN_DB[0..7]
74LS174
hex flip flop
Mr
MAIN_DB1
MAIN_DB0
SCRPOS1D
SCRPOS1
VCC
A8
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
MAIN_DB2
MAIN_DB3
SCRPOS4
SCRPOS5
SCRPOS6
SCRPOS7
SCRPOS0D
SCRPOS0
flip flops
74LS273
CLK_SCOLM
C15E
CLK_SCOLM
Horizontal
Counnter
HF'[0..9]
C4
SCRPOS8
SCRPOS9
HF'9
HF'8
HF'4
HF'5
HF'6
HF'7
HF'0
HF'1
HF'2
HF'3
CO
A1
A2
A3
A4
B1
B2
B3
B4
S1
S2
S3
S4
74LS283
A10
A11
A12
SUM
C93
221
PXLCLKD
PXLCLK
REPAIR
CARRY0
A9
NOT ORIGINALLY CONNECTED
CONNECTION REPAIRED
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
SD7[7..0]
B12
SCR_CA0
SCR_CA1
SCR_CA2
SCR_CA3
SCROM_A4
SCR_CA5
SCR_CA6
SCR_CA7
B10
REPAIR
CARRY0
VF'5
VF'4
B11
SCR_A4
SCR_A5
SCR_A7
SCR_A6
SCR_A8
SCR_A9
B8
RAM_2016 C7
upper half
WE_MAIN
MAIN_AB10
WE_SCR
CE_SCR
RD_SND
F7
VF'6
VF'7
E8E
SCR_AD8
SCR_AD9
SCR_AD10
SCR_AD11
SCR_AD12
SCR_AD13
SCR_AD14
SCR_AD15
SCR0
SCR1
SCR2
SCR3
SCR4
SCR5
SCR6
SCR7
D7
D10
SCR[15..0]
SCROM_A[15..0]
SCROM_A13
SCROM_A12
SCROM_A11
SCROM_A10
SCROM_A9
SCROM_A8
SCROM_A7
SCROM_A6
SCR8
SCR9
SCR10
SCR13
SCR14
SCR15
SCR15
SCR14
SCR13
SCR10
SCR9
SCR8
SCOL0
SCOL1
SCOL2
CE_SCROM
SCROM_A14
SCROM_A15
D8
D9
F8
SCRPXL3
SCRPXL4
SCRPXL5
SCRPXL[5..3]
VF'[0..7]
scrom_memories
SCROM_A[15..0]
PXLCLK
MAIN_DB0
FLIP_LATCH
H0
VF'[0..3]
CE_SCROM
SCR_CA5
SCR_CA6
SCR_CA7
SCRPXL[2..0]
COUNTER_S1D
COUNTER_S0D
CLK_SCOLMD
FLIP_BD
COUNTER_S1
COUNTER_S0
CLK_SCOLM
FLIP_B
COUNTER_S[0..1]
File: scrom_memories.kicad_sch
For repair and maintenance
JOTEGO
Sheet: /scroll/
File: scroll.kicad_sch
Title: RENEGADE
Size: A3
Date: 2022-07-08
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 5/16

SCROM_A[15..0]

SCR_CA5 — C15F 74LS04 — SCROM_A5

CE_SCROM — C15A 74LS04 — CE_SCRB

D11A 74LS139 decoder
- CE_SCROM, SCROM_A14 → A1, A0
- VSS → E
- O0 CE_SCRF
- O1 CE_SCRD
- O2 CE_SCRE
- O3 CE_SCRC

VF'[0..3]

A15 27256 — VCC, VSS, VPP=VCC, CE=CE_SCROM, OE=GND
- A0..A14: SCROM_A4, SCROM_A5, SCROM_A6, SCROM_A7, SCROM_A8, SCROM_A9, SCROM_A10, SCROM_A11, SCROM_A12, SCROM_A13, SCROM_A14
- D0..D7: SCROM_D0, SCROM_D1, SCROM_D2, SCROM_D3, SCROM_D4, SCROM_D5, SCROM_D6, SCROM_D7

B15 27256 — VCC, VSS, VPP=VCC, CE=CE_SCRB, OE=GND
- A0..A14: SCROM_A4, SCROM_A5, SCROM_A6, SCROM_A7, SCROM_A8, SCROM_A9, SCROM_A10, SCROM_A11, SCROM_A12, SCROM_A13, SCROM_A14
- D0..D7: SCROM_D0, SCROM_D1, SCROM_D2, SCROM_D3, SCROM_D4, SCROM_D5, SCROM_D6, SCROM_D7

SCROM_D[0..15]

74LS157 Quad Mux (S, E: VSS)
- I0a SCROM_D0, I0b SCROM_D1, I0c SCROM_D3, I0d SCROM_D2
- I1a SCROM_D4, I1b SCROM_D5, I1c SCROM_D7, I1d SCROM_D6
- A13: SCROM_A15
- Za 4, Zb 7, Zc 12, Zd 9

C14 27256 — VCC, VSS, VPP=VCC, CE=CE_SCRC, OE=GND
- A0..A14: SCROM_A4, SCROM_A5, SCROM_A6, SCROM_A7, SCROM_A8, SCROM_A9, SCROM_A10, SCROM_A11, SCROM_A12, SCROM_A13, SCROM_A15
- D0..D7: SCROM_D8, SCROM_D9, SCROM_D10, SCROM_D11, SCROM_D12, SCROM_D13, SCROM_D14, SCROM_D15

C11 27256 — VCC, VSS, VPP=VCC, CE=CE_SCRD, OE=GND
- A0..A14: SCROM_A4, SCROM_A5, SCROM_A6, SCROM_A7, SCROM_A8, SCROM_A9, SCROM_A10, SCROM_A11, SCROM_A12, SCROM_A13, SCROM_A15
- D0..D7: SCROM_D8, SCROM_D9, SCROM_D10, SCROM_D11, SCROM_D12, SCROM_D13, SCROM_D14, SCROM_D15

C12 27256 — VCC, VSS, VPP=VCC, CE=CE_SCRE, OE=GND
- A0..A14: SCROM_A4, SCROM_A5, SCROM_A6, SCROM_A7, SCROM_A8, SCROM_A9, SCROM_A10, SCROM_A11, SCROM_A12, SCROM_A13, SCROM_A15
- D0..D7: SCROM_D8, SCROM_D9, SCROM_D10, SCROM_D11, SCROM_D12, SCROM_D13, SCROM_D14, SCROM_D15

C9 27256 — VCC, VSS, VPP=VCC, CE=CE_SCRF, OE=GND
- A0..A14: SCROM_A4, SCROM_A5, SCROM_A6, SCROM_A7, SCROM_A8, SCROM_A9, SCROM_A10, SCROM_A11, SCROM_A12, SCROM_A13, SCROM_A15
- D0..D7: SCROM_D8, SCROM_D9, SCROM_D10, SCROM_D11, SCROM_D12, SCROM_D13, SCROM_D14, SCROM_D15

74LS194 shift register — PXLCLK, VCC, SR0-1, SR0-0
- B13: P3 SCROM_D15, P2 SCROM_D14, P1 SCROM_D13, P0 SCROM_D12
- E14: P3 SCROM_D11, P2 SCROM_D10, P1 SCROM_D9, P0 SCROM_D8
- Dsr, Dsl: VSS

74LS157 Quad Mux (S, E: VSS)
- I0a SCROM_D15, I0b SCROM_D14, I0c SCROM_D13, I0d SCROM_D12
- I1a SCROM_D11, I1b SCROM_D10, I1c SCROM_D9, I1d SCROM_D8
- E15: Za SCRPXL1, Zb SCRPXL0, Zc SCRPXL2 → SCRPXL[2..0]

74LS157 Quad Mux (S: FLIP, E: VSS)
- I0a SCROM_D15, I0b SCROM_D14, I0c SCROM_D13, I0d SCROM_D12
- I1a SCROM_D11, I1b SCROM_D10, I1c SCROM_D9, I1d SCROM_D8
- D15: Za COUNTER_S0 SR0-0, Zb COUNTER_S1 SR0-1, Zc, Zd

SCR_CA6 — D13B 74LS86 — F14A 74LS08 — H0
SCR_CA7 — D13A 74LS86 — F14A 74LS08

G15A 74LS74 (D: MAIN_DB0, C: FLIP_LATCH, Q → FLIP_B via 74LS04)

FLIP

SCROM_A4 — D13D 74LS86 — CLK_SCOLM

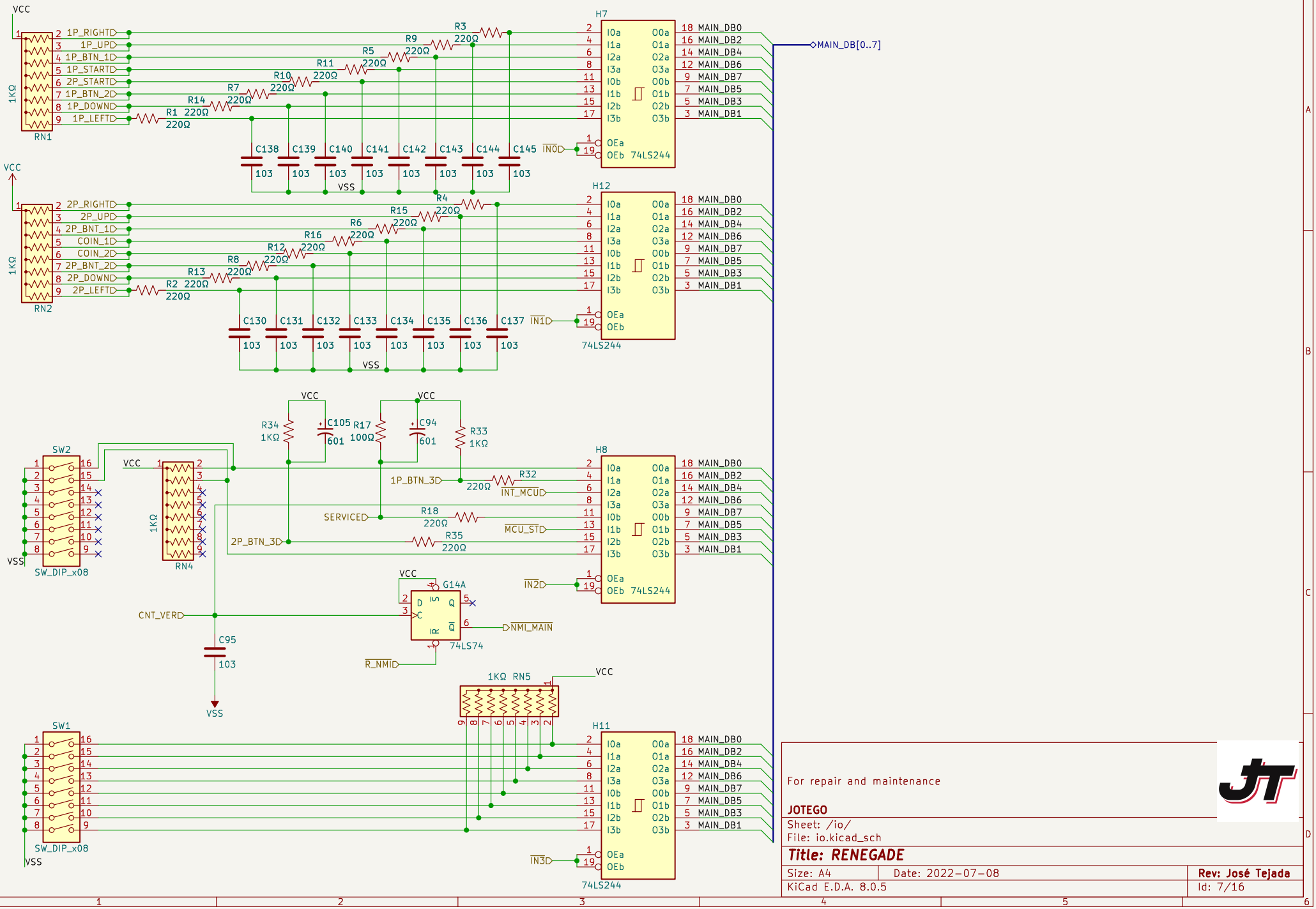
For repair and maintenance

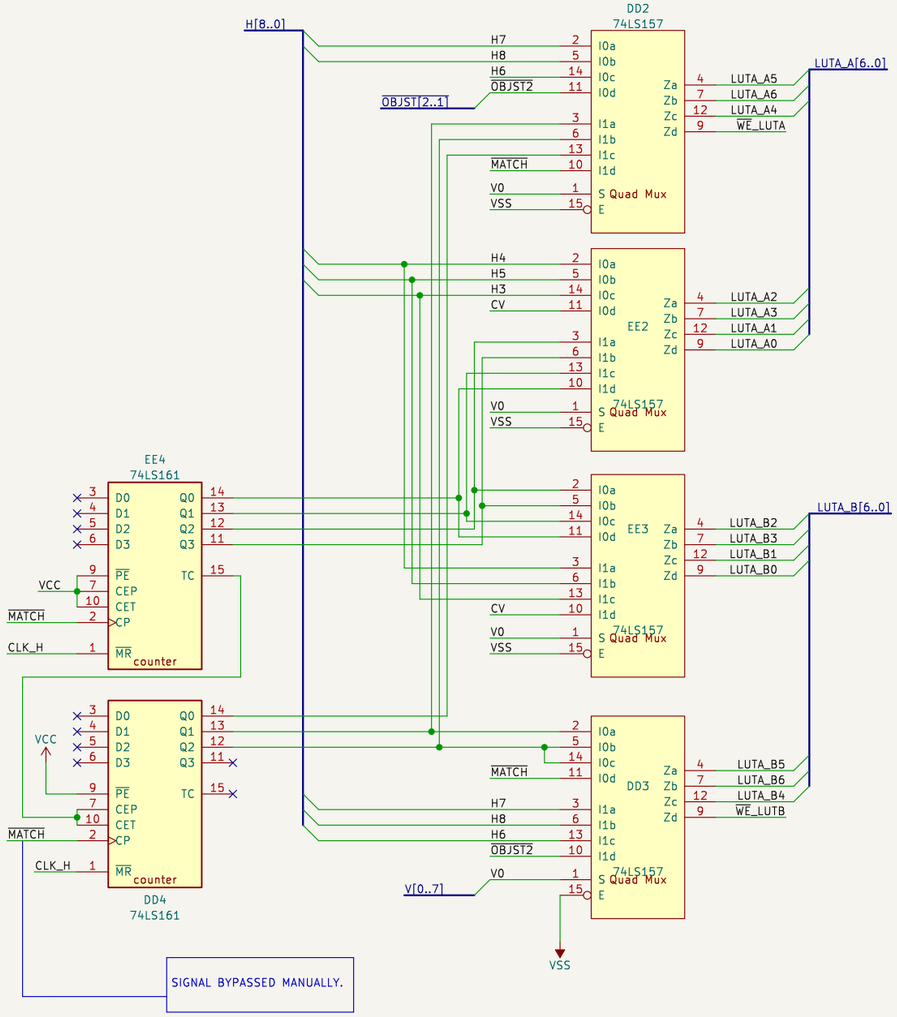
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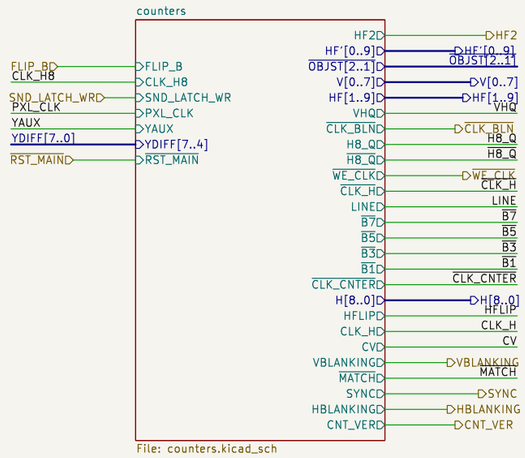
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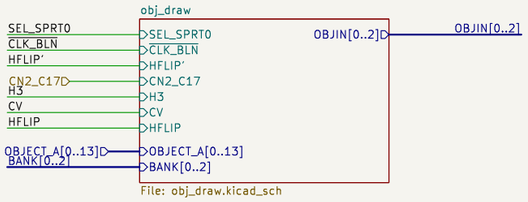
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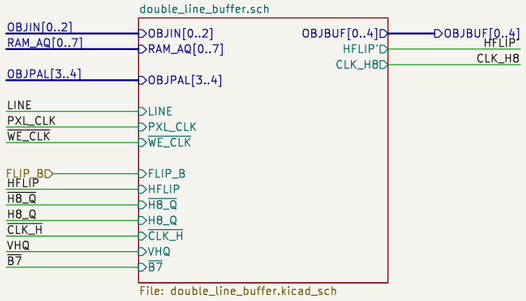
Size: A3	Date: 2022-07-08	Rev: José Tejada
KiCad E.D.A. 8.0.5		Id: 6/16

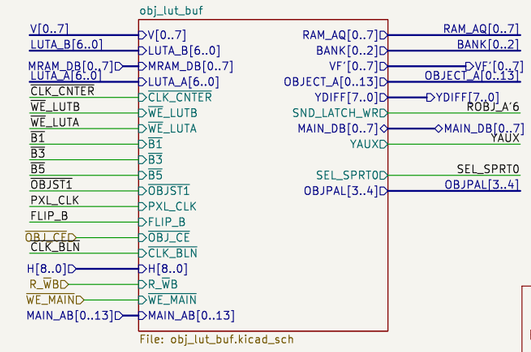

VCC
1P_RIGHT
1P_UP
1P_BTN_1D
1P_START
2P_START
1P_BTN_2D
1P_DOWN
1P_LEFT
RN1
1KΩ
R1 220Ω
R2 220Ω
R3 220Ω
R4 220Ω
R5 220Ω
R6 220Ω
R7 220Ω
R8 220Ω
R9 220Ω
R10 220Ω
R11 220Ω
R12 220Ω
R13 220Ω
R14 220Ω
R15 220Ω
R16 220Ω
C138 C139 C140 C141 C142 C143 C144 C145
C130 C131 C132 C133 C134 C135 C136 C137
103
VSS
IN0D
IN1D
H7
H12
74LS244
MAIN_DB[0..7]
MAIN_DB0
MAIN_DB2
MAIN_DB4
MAIN_DB6
MAIN_DB7
MAIN_DB5
MAIN_DB3
MAIN_DB1
2P_RIGHT
2P_UP
2P_BNT_1D
COIN_1D
COIN_2D
2P_BNT_2D
2P_DOWN
2P_LEFT
RN2
SW2
SW_DIP_x08
RN4
R34 1KΩ
R17 100Ω
R33 1KΩ
C105 601
C94 601
R32 220Ω
R18 220Ω
R35 220Ω
1P_BTN_3D
2P_BTN_3D
SERVICED
INT_MCU
MCU_ST
IN2D
H8
G14A
74LS74
CNT_VER
NMI_MAIN
R_NMI
C95 103
1KΩ RN5
SW1
IN3D
H11
For repair and maintenance
JOTEGO
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Title: RENEGADE
Size: A4
Date: 2022-07-08
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 7/16











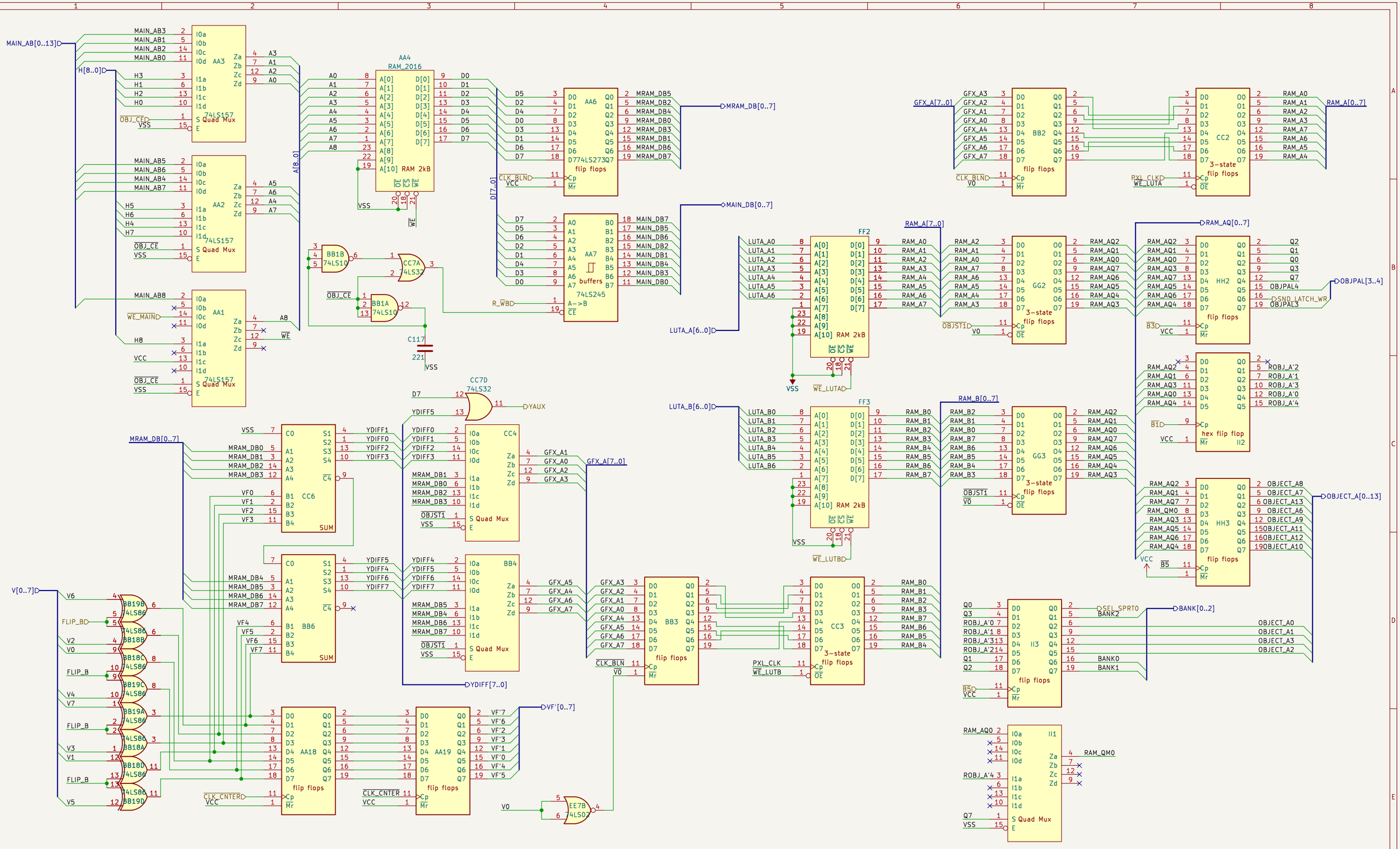
For repair and maintenance

JOTEGO

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Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 8/16



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JOTEGO

Sheet: /objects/obj_lut_buf/
File: obj_lut_buf.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 9/16

VERTICAL COUNTER

HORIZONTAL COUNTER

OBJECT STATE MACHINE

SHORTED, BUT NOT CONNECTED

SIGNAL BYPASSED MANUALLY.

REPAIR

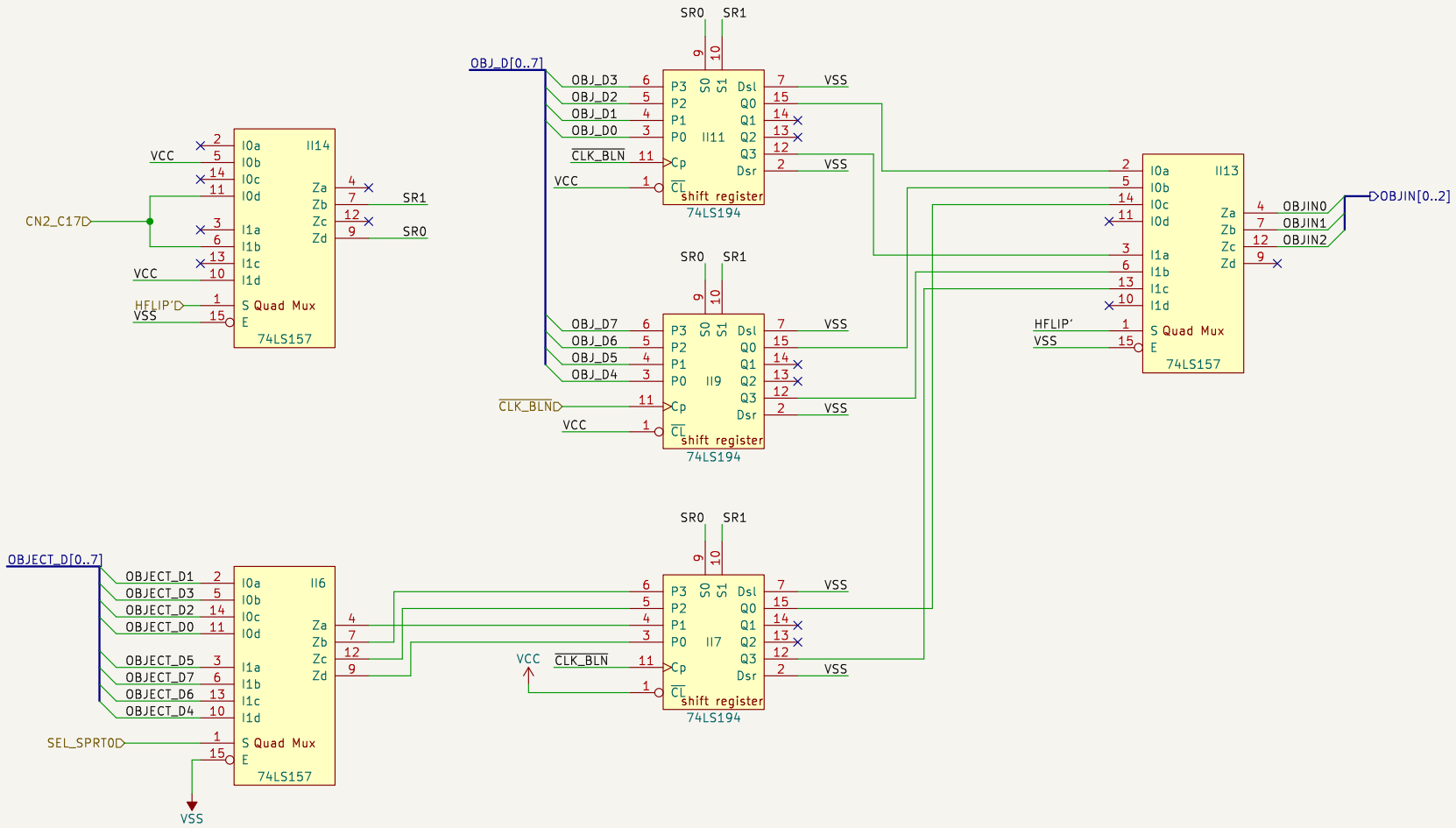
For repair and maintenance

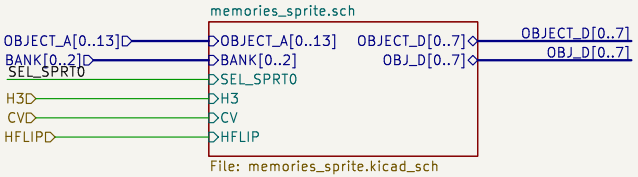
JOTEGO

Sheet: /objects/counters/
File: counters.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
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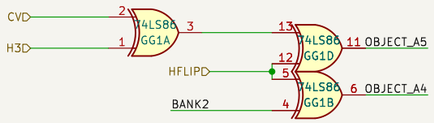
For repair and maintenance

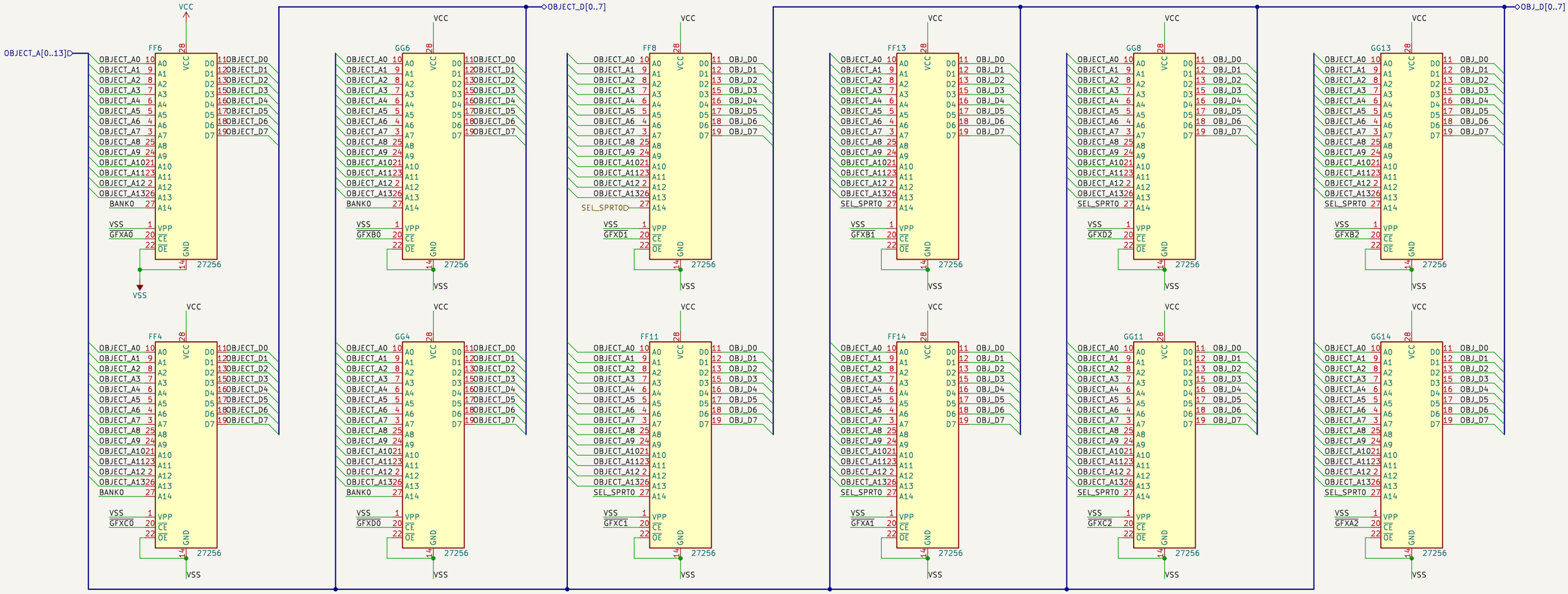
JOTEGO

Sheet: /objects/obj_draw/
File: obj_draw.kicad_sch

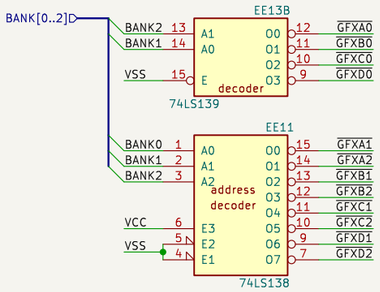
Title: RENEGADE

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Memorias volcadas



For repair and maintenance

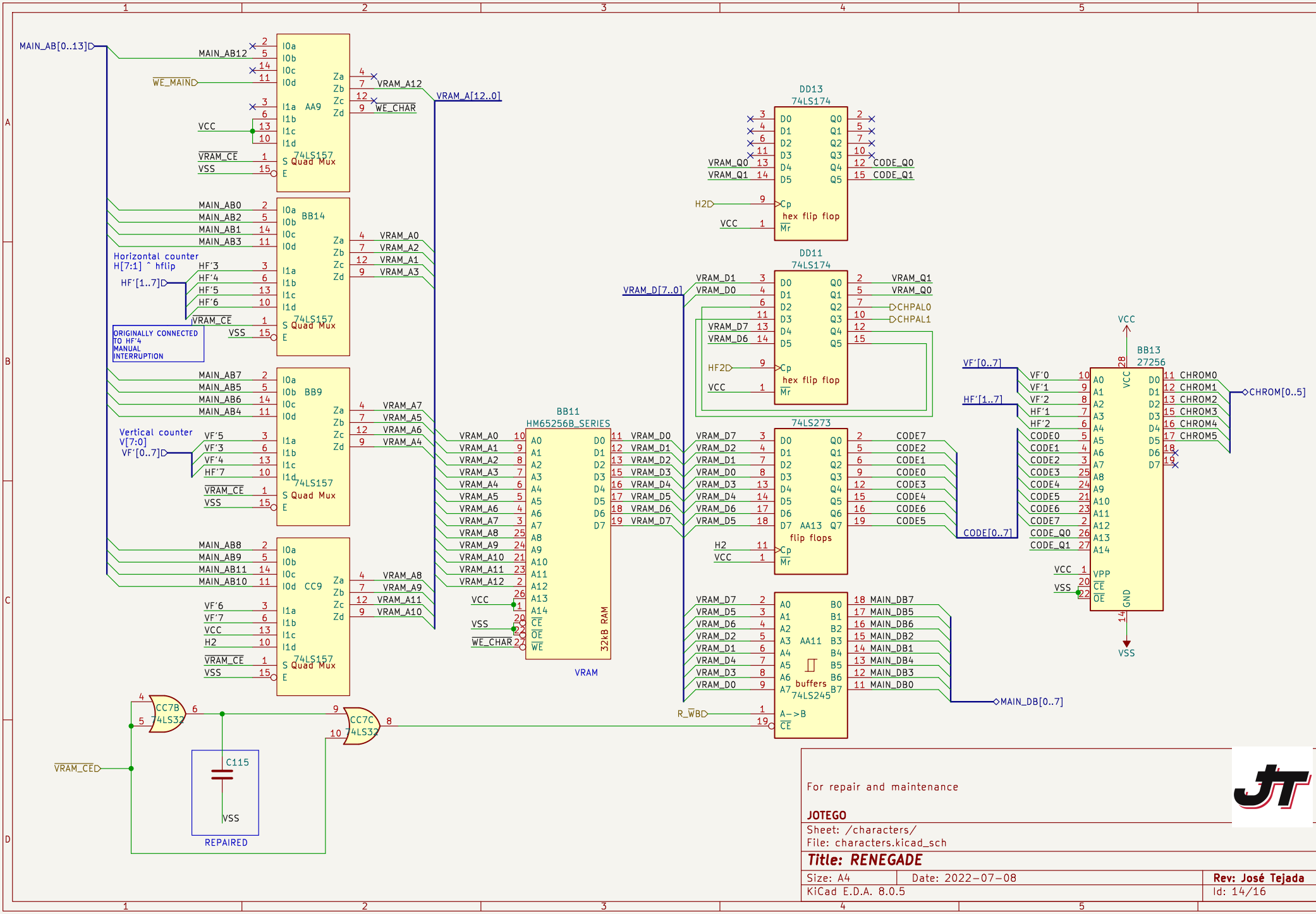
JOTEGO

Sheet: /objects/obj_draw/memories_sprite.sch/
File: memories_sprite.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
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RAM_AQ[0..7]
OBJIN[0..2]
LINEA[4..0]
LINEB[4..0]
OBJPAL[3..4]
OBJBUF[0..4]
HFLIP
VSS
VCC
WE_CLK
PXL_CLK
LINE
BLANK
BUFIN0
BUFIN1
BUFIN2
BUFIN3
BUFIN4
LINEA0
LINEA1
LINEA2
LINEA3
LINEA4
LINEB0
LINEB1
LINEB2
LINEB3
LINEB4
OBJBUF0
OBJBUF1
OBJBUF2
OBJBUF3
OBJBUF4
HFLIP'
BLANK'
FLIP_BD
CLK_H8
H8_Q
B7
VHQ
REPAIRED
For repair and maintenance
JOTEGO
Sheet: /objects/double_line_buffer.sch/
File: double_line_buffer.kicad_sch
Title: RENEGADE
Size: A3
Date: 2022-07-08
Rev: José Tejada
KiCad E.D.A. 8.0.5
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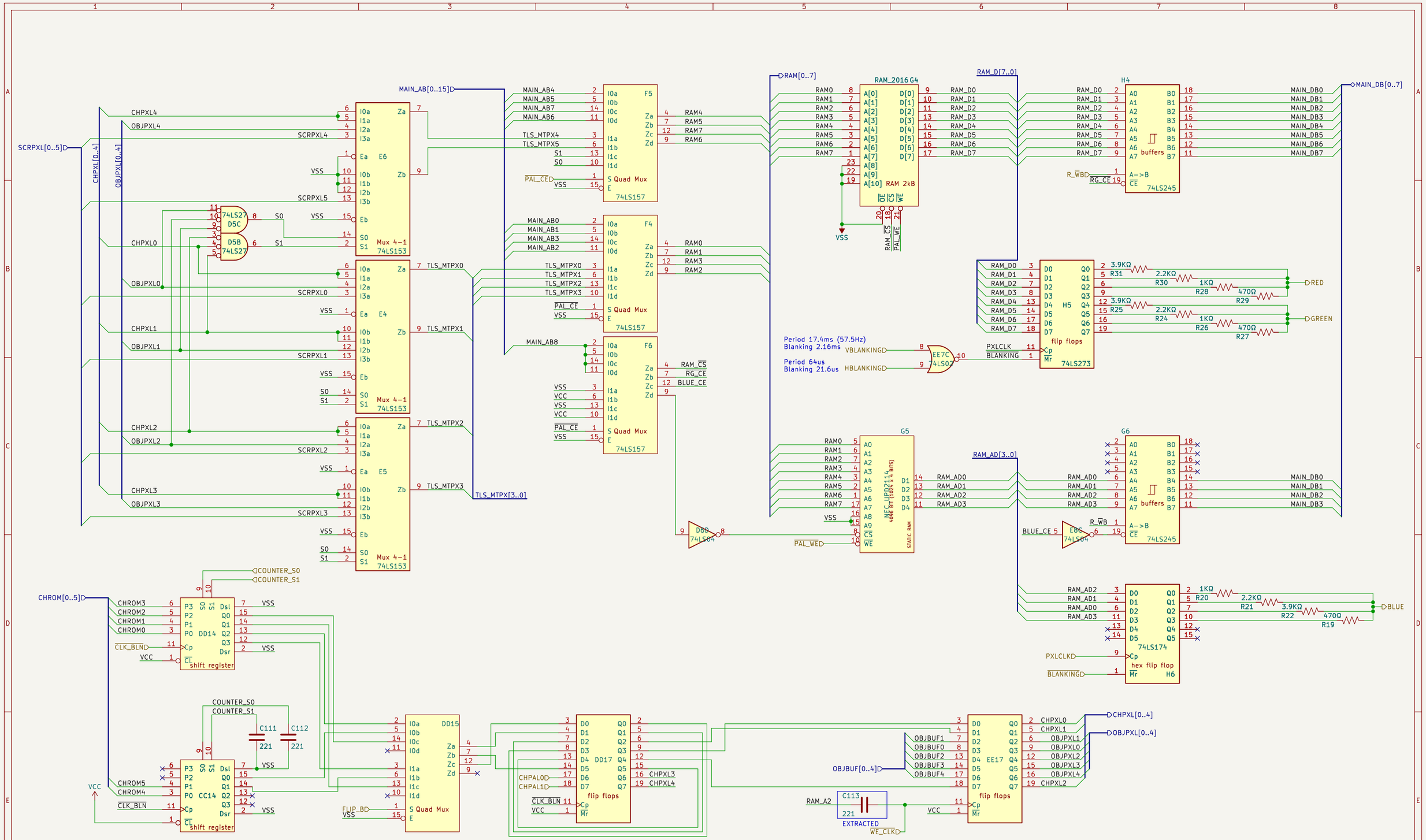
JOTEGO

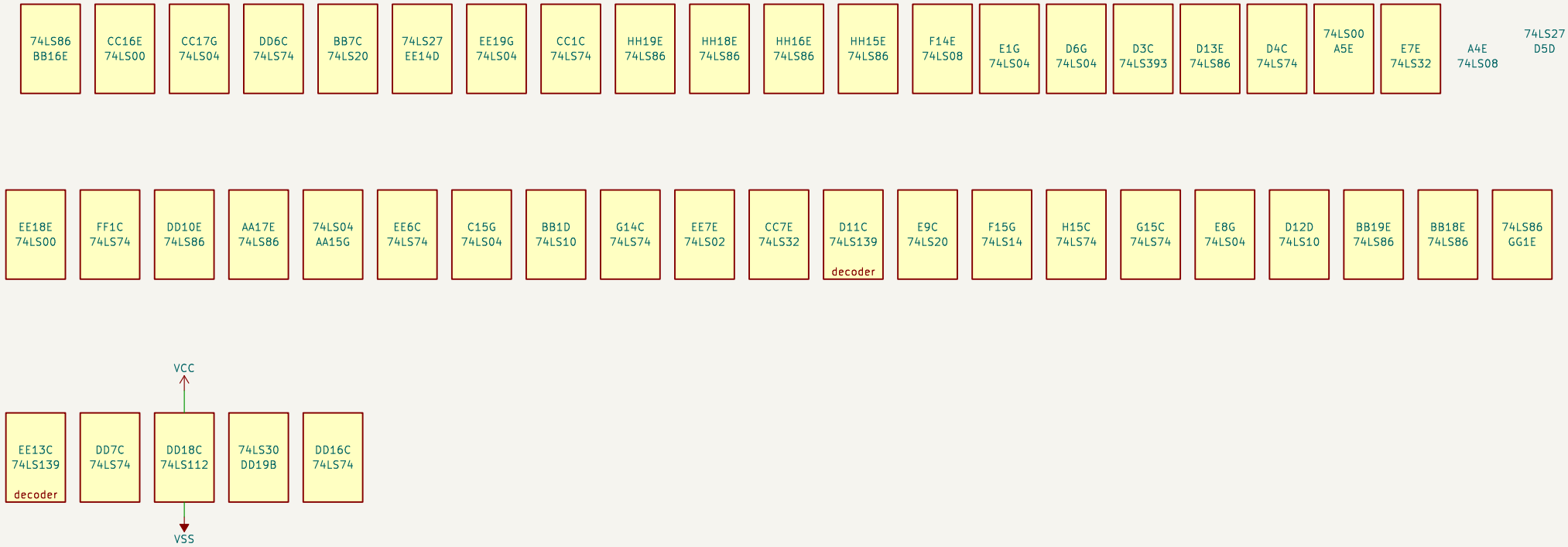
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File: characters.kicad_sch

Title: RENEGADE

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MAIN_AB[0..15]
SCRPXL[0..5]
CHPXL[0..4]
OBJPXL[0..4]
CHPXL4
OBJPXL4
SCRPXL4
VSS
SCRPXL5
74LS27
D5C
D5B
S0
S1
CHPXL0
OBJPXL0
SCRPXL0
CHPXL1
OBJPXL1
SCRPXL1
CHPXL2
OBJPXL2
SCRPXL2
CHPXL3
OBJPXL3
SCRPXL3
E6
E4
E5
Mux 4–1
74LS153
TLS_MTPX0
TLS_MTPX1
TLS_MTPX2
TLS_MTPX3
TLS_MTPX[3..0]
MAIN_AB4
MAIN_AB5
MAIN_AB7
MAIN_AB6
TLS_MTPX4
TLS_MTPX5
PAL_CE
74LS157
RAM4
RAM5
RAM7
RAM6
MAIN_AB0
MAIN_AB1
MAIN_AB3
MAIN_AB2
RAM0
RAM1
RAM3
RAM2
MAIN_AB8
RAM_CS
RG_CE
BLUE_CE
VCC
RAM[0..7]
RAM_2016 G4
A[10] RAM 2kB
OE
CS
WE
RAM_CS
PAL_WE
RAM_D[7..0]
RAM_D0
RAM_D1
RAM_D2
RAM_D3
RAM_D4
RAM_D5
RAM_D6
RAM_D7
H4
buffers
74LS245
R_WB
MAIN_DB[0..7]
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
Period 17.4ms (57.5Hz)
Blanking 2.16ms
Period 64us
Blanking 21.6us
VBLANKING
HBLANKING
EE7C
74LS02
H5
flip flops
74LS273
PXLCLK
BLANKING
3.9KΩ
2.2KΩ
1KΩ
470Ω
R31
R30
R28
R29
R25
R24
R26
R27
RED
GREEN
G5
NEC UPD2114
4096 BIT (1024 x 4 BITS)
STATIC RAM
RAM_AD[3..0]
RAM_AD0
RAM_AD1
RAM_AD2
RAM_AD3
G6
EE6C
74LS04
R20
R21
R22
R19
BLUE
74LS174
hex flip flop
H6
COUNTER_S0
COUNTER_S1
CHROM[0..5]
CHROM3
CHROM2
CHROM1
CHROM0
CLK_BLN
shift register
C111
C112
221
CHROM5
CHROM4
CC14
DD14
DD15
DD17
EE17
FLIP_BD
CHPAL0
CHPAL1
CHPXL3
CHPXL4
OBJBUF[0..4]
OBJBUF1
OBJBUF0
OBJBUF2
OBJBUF3
OBJBUF4
CHPXL2
OBJPXL1
OBJPXL2
OBJPXL3
OBJPXL4
RAM_A2
C113
EXTRACTED
WE_CLK
For repair and maintenance
JOTEGO
Sheet: /colmix/
File: colmix.kicad_sch
Title: RENEGADE
Size: A3
Date: 2022–07–08
KiCad E.D.A. 8.0.5
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JOTEGO

Sheet: /last unit/
File: last unit.kicad_sch

Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
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